MEMORANDUM

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RE: **Flat Plate Boundary Layer Flow**

Introduction

In this lab, the boundary layer for flow over a flat plate was characterized using pressure measurements from a Pitot-static tube at different distances from the flat plate. This data was then used to calculate boundary layer, displacement, and momentum thicknesses, drag, and drag coefficients using numerical integration techniques. To label the flow as laminar, turbulent, or transitional, the experimental data was compared to values obtained through analytical modeling. The measured boundary layer thickness was 1.645 mm, with momentum and displacement thicknesses of 0.167 mm and 0.368 mm, respectively. The boundary layer thickness for the laminar predictions is 1.614 mm and 5.038 mm using turbulent flow correlations. The calculated drag force was 0.0335 N with a drag coefficient of 0.00202. Comparison with theoretical predictions for laminar and turbulent flow indicates that the flow was laminar at the measurement location.

Background

When a uniform free-stream flow passes over a flat plate, the no-slip condition requires the fluid velocity at the surface to be zero relative to the surface, creating what is known as a boundary layer. Within the boundary layer, velocities are characterized by high gradients, viscous effects dominate, and shear stresses develop, while the flow outside the boundary layer can be modeled as inviscid. Figure 1 displays a two-dimensional boundary layer formed by constant fluid velocity parallel to a flat plate.

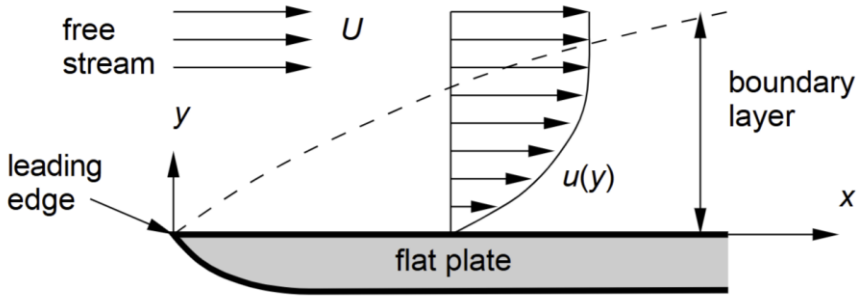


Figure 1. Boundary layer for uniform flow over a flat plate.

Boundary layer behavior over a flat plate is characterized by the Reynolds number based on the distance from the leading edge,

$$Re_x = \frac{Ux}{\nu} \tag{1}$$

where U is the free-stream velocity, x is the distance from the leading edge of the plate, and ν is the kinematic viscosity of the fluid. This parameter determines whether the boundary layer is laminar, transitional, or turbulent. For uniform flow over a smooth flat plate, a laminar boundary layer forms at the leading edge and persists until approximately

$$Re_x \approx 5 \times 10^5$$

at which point transition to turbulence may begin. The exact transition location depends on surface roughness and free-stream disturbances.

Three parameters are commonly used to characterize boundary layer growth: the boundary layer thickness, δ, the displacement thickness, δ^*, and the momentum thickness, Θ. The boundary layer thickness is defined as the location where the velocity reaches 99% of the free-stream velocity. The displacement thickness represents the effective reduction in mass flow due to the presence of the boundary layer, while the momentum thickness represents the reduction in momentum flux caused by viscous effects.

For incompressible flow, these integral thicknesses are defined as:

$$\delta^* = \int_0^\infty \left(1 - \frac{u}{U}\right) dy \tag{2}$$

$$\Theta = \int_0^\infty \frac{u}{U}\left(1 - \frac{u}{U}\right) dy \tag{3}$$

The momentum thickness is directly related to the drag force acting on the plate through the von Kármán momentum integral relation:

$$F_D = \rho b U^2 \Theta \tag{4}$$

where b is the width of the plate perpendicular to the free-stream flow.

The dimensionless drag coefficient can then be expressed as:

$$C_D = \frac{F_D}{\frac{1}{2}\rho b L U^2} = \frac{2\Theta}{L} \tag{5}$$

where L is the streamwise distance from the leading edge to the location where Θ is evaluated.

Laminar Boundary Layer Model

For laminar flow over a flat plate with zero pressure gradient, the velocity profile can be described by the Blasius solution to the boundary layer equations. Equation 6 describes how the laminar boundary layer thickness grows.

$$\delta = \frac{5.0x}{\sqrt{Re_x}} \tag{6}$$

Equation 7 can be used to determine the displacement thickness for laminar flow and Equation 8 is used to calculate the momentum thickness.

$$\delta^* = \frac{1.721x}{\sqrt{Re_x}} \tag{7}$$

$$\Theta = \frac{0.664x}{\sqrt{Re_x}} \tag{8}$$

Turbulent Boundary Layer Model

For turbulent boundary layer flow, an analytical solution is not available, and empirical correlations are used. A widely used approximation for the turbulent velocity distribution is the power-law profile:

$$\frac{u(y)}{U} = \left(\frac{y}{\delta}\right)^{\frac{1}{n}}, 0 \le y \le \delta \tag{9}$$

where $n \approx 7$for most turbulent boundary layers.

For turbulent flow over a flat plate, the boundary layer thickness grows more rapidly than in laminar flow and scales approximately as:

$$\delta = \frac{0.370x}{Re_x^{1/5}} \tag{10}$$

The correlations used to determine the displacement thickness and momentum thickness for a turbulent boundary layer are shown in Equations (11) and (12).

$$\delta^* = \frac{0.0463x}{Re_x^{1/5}} \tag{11}$$

$$\Theta = \frac{0.0360x}{Re_x^{1/5}} \tag{12}$$

This framework allows experimental boundary layer thickness, displacement thickness, momentum thickness, and drag coefficient to be compared with laminar and turbulent predictions to determine the flow regime at the measurement location.

Procedure/Analysis

Prior to the experiment, the temperature and pressure in the room were recorded. The equipment used for the experiment is shown in Figure 2. The pressure tap located in the larger cross-sectional area portion of the apparatus was connected to a multi-tube manometer. The multi-tube manometer was inclined to 60 degrees for better resolution. This pressure measurement was used to find the free-stream velocity in the test section. A flat plate with a smooth side was installed and set to a height of 16.5 cm in the test section using a set screw.



Figure 2. Boundary Layer Experiment Setup

The fan was turned on and the damper was opened to about 45 degrees. A pitot-static tube, connected to an inclined manometer, was moved towards the flat plate using a micrometer. A small piece of plastic with a thickness of 0.25 mm was placed between the pitot-static tube and the flat plate. The first measurement

with the pitot-static tube was recorded. The pitot-static tube was moved 0.05 mm away from the flat plate and another measurement was recorded. This process was repeated until the pitot-static tube was located 0.70 mm away from the flat plate surface. The pitot-static tube was moved in 0.10 mm increments until 1.00 mm and 0.20 mm increments until 1.60 mm. The final two measurements were recorded at 1.90 mm and 2.20 mm. Measurements were initially taken at smaller measurements because the boundary layer velocity was changing more rapidly. As the pitot-static tube is moved closer to the boundary layer thickness, the velocity approaches the free-stream velocity and does not change as rapidly.

The pressure measurements were used to find the velocity at each point measured in the boundary later and in the free stream. Using the experimental data collected, the boundary layer thickness, δ, displacement thickness, δ^*, and momentum thickness, Θ , were calculated. Specifically, the boundary layer thickness was calculated by linearly interpolating the data to find the location where 99% of the free stream velocity was reached. The displacement thickness was calculated using Equation 3 and the momentum thickness was calculated using Equation 4. The trapezoidal rule was used to numerically integrate the quantities in these equations to find the final value for the displacement and momentum thickness.

To compare experimental results to theoretical results, the Blasius solution for laminar flow and turbulent flow correlations were used to predict the boundary layer, displacement, and momentum thicknesses. Equations (6), (7), and (8) were used to find these values for laminar flow, while Equations (9), (10), and (11) were used to find the thicknesses for turbulent flow. The drag coefficient was calculated using Equation 5.

Results and Discussion

The data collected during the experiment and the streamwise velocity is shown in Attachment A. From the experimental data, the velocity at each y-location from the flat plate was calculated and normalized by the free-stream velocity. The normalized free-stream velocity values were used to find the location of where the velocity in the boundary layer, *u*, reaches 99% of the free-stream velocity, *U.* As shown in Table 1, the calculated boundary layer thickness is 1.65 mm. The terms that were numerically integrated to calculate the momentum and displacement thickness are also shown in Table 1. The calculated value for the momentum thickness is 0.167 mm and the displacement thickness is 0.368 mm. Sample calculations for the displacement and momentum thicknesses are shown in Attachment B.

Table 1. Normalized Velocity, Boundary Layer Thickness, Momentum Thickness, and Displacement Thickness

Normalized Distance from Plate, y/δ	Normalized Streamwise Velocity, $\frac{u}{U}$	$1-\frac{u}{U}$	$\frac{u}{U}(1-\frac{u}{U})$	Partial Integrals (Momentum Thickness, Θ)	Partial Integrals (Displacement Thickness, δ^*)
0.15	0.481	0.519	0.250	0.0312	0.1899
0.18	0.540	0.460	0.248	0.0125	0.0245
0.21	0.641	0.359	0.230	0.0120	0.0205
0.24	0.686	0.314	0.215	0.0111	0.0168
0.27	0.733	0.267	0.196	0.0103	0.0145
0.30	0.778	0.222	0.173	0.0092	0.0122
0.33	0.806	0.194	0.156	0.0082	0.0104
0.36	0.851	0.149	0.127	0.0071	0.0086
0.40	0.873	0.127	0.111	0.0059	0.0069
0.43	0.894	0.106	0.095	0.0051	0.0058
0.49	0.894	0.106	0.095	0.0095	0.0106
0.55	0.915	0.085	0.078	0.0086	0.0096
0.61	0.935	0.065	0.061	0.0070	0.0075
0.73	0.955	0.045	0.043	0.0104	0.0111
0.85	0.974	0.026	0.025	0.0069	0.0072
0.97	0.974	0.026	0.025	0.0051	0.0052
1.16	0.993	0.007	0.007	0.0049	0.0050
1.34	0.993	0.007	0.007	0.0021	0.0021
	δ = 1.65 mm			**Θ = 0.167 mm**	**δ^* = 0.368 mm**

From the free-stream velocity, the Reynolds number for the flow was calculated using Equation 1. The Reynolds number for the experiment was 2.61 x 10^5, which would generally classify the flow as laminar.

Using this Reynolds number, the theoretical values for the boundary layer, displacement, and momentum thicknesses were computed using the Blasius solution for laminar flow and empirically determined correlations for turbulent flow. Table 2 shows a comparison of these theoretical values and the values determined from the experiment. Sample calculations for the thicknesses for the laminar and turbulent predictions are shown in Attachment C. Additionally, Table 2 shows the calculated drag coefficient for the experimental results and the laminar and turbulent flow predictions. Sample calculations for the drag coefficient are shown in Attachment D.

Table 2. Comparison of Experimental Results and Predictions for Boundary Layer, Displacement, and Momentum Thicknesses

	Boundary Layer Thickness, δ [mm]	Displacement Thickness, δ^* [mm]	Momentum Thickness, Θ [mm]	Drag Coefficient, C_D
Experiment	1.645	0.368	0.167	0.0020
Laminar Prediction	1.614	0.556	0.214	0.0026
Turbulent Prediction	5.038	0.630	0.490	0.0058

As shown in Table 2, the experimentally determined value of 1.645 mm for the boundary layer thickness is very close to the laminar prediction of 1.614 mm. The experimental results for the displacement and momentum thickness are much smaller than the theoretical values. However, they are closer to the laminar prediction values, confirming that the flow from the experiment more closely resembled laminar flow. The smaller values for the displacement and momentum thicknesses for the experimental setup indicate that the boundary layer had less momentum and mass flux compared to the theoretical model. This could have been due to the fact that the thickness of the pitot-static tube prevented the ability to take measurements as close as possible to the surface of the flat plate. There would be the most momentum and mass flux losses at the surface of the flat plate, so this could cause the experimental values to seem lower.

The drag coefficients for the experimental results and laminar and turbulent predictions support the conclusion that the experiment reflected less momentum and mass flux losses than the predictions indicate. The drag coefficient for the experimental results is the lowest, indicating less resistance over the plate compared to the predictions. Additionally, the drag coefficient is highest for the turbulent prediction over the flat plate. This is because turbulence increases the skin friction, which results in a higher friction drag. The drag for the experimental results was calculated to be 0.0334 N. Overall, this value seems small, but reasonable. This is because there is only friction due to drag contributing to this value because a flat plate was used.

Figures 3a and 3b show the comparison between the theoretical laminar, turbulent, and experimental results for both dimensional and dimensionless data. The laminar profile in Figure 3a increases gradually and exhibits parabolic growth associated with a thicker boundary layer and lower near-wall velocity gradients. The turbulent profile rises steeply near the wall, which is expected from enhanced momentum transport and fuller velocity profile. The experimental data for Figure 3a initially increases more gradually than the turbulent curve but much more rapidly than the laminar prediction. Beyond mid-boundary-layer thickness, the experimental values closely approach the turbulent curve and reach nearly the same free-stream velocity at the edge of the boundary layer.

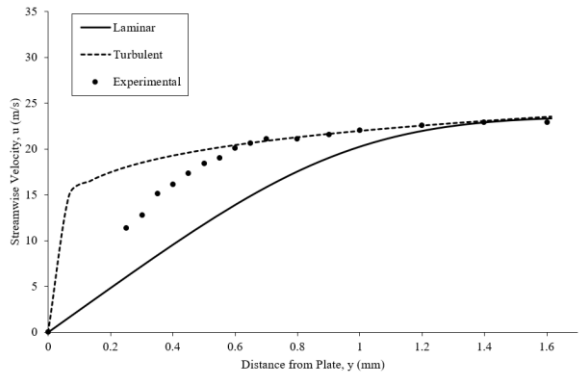


Figure 3a. Streamwise velocity, u vs. Distance from the Plate, y for Experimental Data and Laminar and Turbulent Predictions

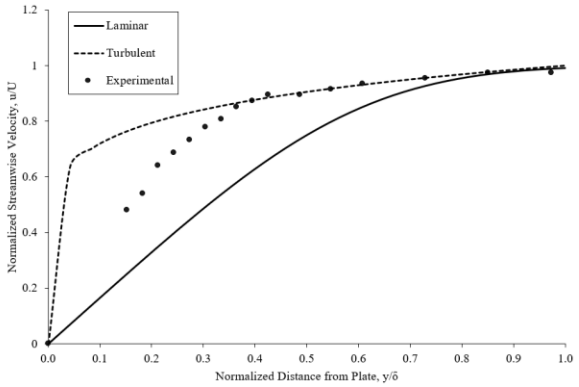


Figure 3b. Normalized Streamwise velocity, u/U vs. Normalized Distance from the Plate, y/δ for Experimental Data and Laminar and Turbulent Predictions

Figure 3b reinforces these conclusions between experimental data and theoretical predictions. The laminar profile exhibits a smooth, concave-down shape consistent with the Blasius solution, while the turbulent power-law profile shows higher velocities near the wall and a fuller boundary layer shape. The experimental data closely follows the laminar prediction throughout most of the boundary layer thickness and deviates significantly from the turbulent profile at lower values of y/δ.

The Reynolds number at the measurement location was calculated as $Re_x = 2.61 \times 10^5$. Since transition for a smooth flat plate typically occurs near $Re_x \approx 5 \times 10^5$, the measured Reynolds number is below the expected transition threshold. However, based on Figure 3a and Figure 3b, the experimental data seems to indicate a transitional flow because the velocity data is between the predicted profiles for the turbulent and laminar flows. If the rough side of the flat plate was used, the flow would have likely followed the velocity profile for turbulent flow more closely. A closer investigation into the other parameters in the lab leads to a more conclusive answer.

A quantitative comparison of boundary layer thickness is in Attachment E. Specifically, the difference between the laminar prediction and experimental result for the boundary layer thickness is 1.9%. The difference between the predicted value for turbulent flow and the experimental data is 67.3%, showing that the flow more closely follows the trends of laminar flow. Similarly, the experimentally determined drag coefficient was significantly closer to the laminar prediction than the turbulent correlation.

Based on the Reynolds number, boundary layer thickness comparison, drag coefficient agreement, and velocity profile shape, the boundary layer at the measurement location is classified as laminar. Minor discrepancies between experimental and theoretical values may be attributed to measurement uncertainty, slight free-stream disturbances, or minor surface imperfections; however, the overall agreement strongly supports laminar behavior.

References

1. ME 347 – Lab 6 Manual, Arturo Cajal.
2. Fox and McDonald's Introduction to Fluid Mechanics, 9th Edition, Philip J. Pritchard and John W. Mitchell.

Attachments

A. Raw Experimental Data and Streamwise Velocity

B. Sample Calculations for Displacement and Momentum Thickness

C. Laminar and Turbulent Predictions for Boundary Layer, Displacement, and Momentum Thickness

D. Sample Calculation for Drag Force and Drag Force Coefficient

E. Quantitative Comparison of Experimental Results to Laminar and Turbulent Predictions

ENGINEERING DATA SHEET	Sheet No.	A – 1/1	Date: 03/07/26
Attachment A: Raw Experimental Data and Streamwise Velocity	Prepared	KG	03/07/26
	Checked	GL	03/07/26
	Approved	GC	03/07/26

Raw data collected from the experiment is shown in Table 2.

Table 2. Experimental Data

Distance from Plate, y [mm]	Pitot-tube Pressure [mm H2O]	Pressure [Pa]	Stream-wise Velocity, u (y) [m/s]
0.25	7.9	77.2	11.33
0.30	9.9	97.1	12.71
0.35	14.0	137	15.10
0.40	16.0	157	16.16
0.45	18.3	179	17.27
0.50	20.6	202	18.32
0.55	22.1	217	18.99
0.60	24.6	242	20.05
0.65	25.9	254	20.56
0.70	27.2	267	21.06
0.80	27.2	267	21.06
0.90	28.4	279	21.54
1.00	29.7	291	22.02
1.20	31.0	304	22.48
1.40	32.3	316	22.94
1.60	32.3	316	22.94
1.90	33.5	329	23.39
2.20	33.5	329	23.39

ENGINEERING DATA SHEET	Sheet No.	B – 1/2	Date: 03/07/26
Attachment B: Sample Calculations for Displacement and Momentum Thickness	Prepared	KG	03/07/26
	Checked	LS	03/07/26
	Approved	GL	03/07/26

The trapezoidal rule was used to numerically integrate the partial integrals for the displacement thickness, δ^*, and momentum thickness, Θ. The equations for the displacement and momentum thicknesses are shown below.

$$\delta^* = \int_0^\infty \left(1 - \frac{u}{U}\right) dy$$

$$\Theta = \int_0^\infty \frac{u}{U}\left(1 - \frac{u}{U}\right) dy$$

Displacement Thickness:

The trapezoidal rule for numerical integration with the appropriate substitution for the first numerical integration is shown below. The free stream velocity was calculated using the relationship between velocity and dynamic pressure.

$$\frac{1}{2}[f(x_i) + f(x_{i+1})]((x_{i+1} - x_i) = \frac{1}{2}[(1 - \frac{u_1}{U}) + (1 - \frac{u_2}{U})]((y_2 - y_1)$$

The streamwise velocity was calculated from the pitot-static tube measurements from the inclined manometer. The pressure measured from the pitot-static tube in mm H_2O was converted to Pascals using the hydrostatic equation. The density of the air was calculated using the ideal gas law. For the first streamwise velocity measurement in the boundary layer,

$$u_1 = \sqrt{\frac{2p_{dyn}}{\rho}} = \sqrt{\frac{2(7.874\ mm\ H_2O)\left(9.81\frac{m}{s^2}\right)\left(\frac{1000kg}{m^3}\right)}{(1000\ mm)\left(1.202\frac{kg}{m^3}\right)}} = 11.3\ m/s$$

For the first partial integral,

$$\tfrac{1}{2}[(1 - \tfrac{11.3}{23.6}) + (1 - \tfrac{12.7}{23.6})]((0.3 - 0.25) = 0.190\ mm.$$

Summing all of the partial integrals for the displacement thickness,

$$\delta^* = \int_0^\infty \left(1 - \frac{u}{U}\right) dy = 0.368\ mm.$$

Momentum Thickness:

The trapezoidal rule for numerical integration with the appropriate substitution for the first numerical integration is shown below. The free stream velocity was calculated using the relationship between velocity and dynamic pressure.

$$\frac{1}{2}[f(x_i) + f(x_{i+1})]((x_{i+1} - x_i) = \frac{1}{2}[\frac{u_1}{U}\left(1 - \frac{u_1}{U}\right) + \frac{u_2}{U}\left(1 - \frac{u_2}{U}\right)]((y_2 - y_1)$$

The streamwise velocity was calculated from the pitot-static tube measurements from the inclined manometer. The pressure measured from the pitot-static tube in mm H_2O was converted to Pascals using the hydrostatic equation. The density of the air was calculated using the ideal gas law. For the first streamwise velocity measurement in the boundary layer,

$$u_1 = \sqrt{\frac{2p_{dyn}}{\rho}} = \sqrt{\frac{2(7.874\ mm\ H_2O)\left(9.81\frac{m}{s^2}\right)\left(\frac{1000kg}{m^3}\right)}{(1000\ mm)\left(1.202\frac{kg}{m^3}\right)}} = 11.3\ m/s$$

For the first partial integral,

$$\frac{1}{2}[\frac{11.3}{23.6}(1 - \frac{11.3}{23.6}) + \frac{11.3}{23.6}(1 - \frac{12.7}{23.6})]((0.3 - 0.25) = 0.0312\ mm.$$

Summing all of the partial integrals for the displacement thickness,

$$\Theta = \int_0^\infty \frac{u}{U}\left(1 - \frac{u}{U}\right) dy = 0.167\ mm.$$

ENGINEERING DATA SHEET	Sheet No.	C – 1/2	Date: 03/07/26
Attachment C: Laminar and Turbulent Predictions for Boundary Layer, Displacement, and Momentum Thickness	Prepared	KG	03/07/26
	Checked	GC	03/07/26
	Approved	GL	03/07/26

The laminar and turbulent thickness predictions are functions of Reynolds number, so the Reynolds number was first calculated:

$$Re_x = \frac{Ux}{\nu} = \frac{(\frac{23.56\ m}{s})(0.165\ m)}{1.79\ x\ 10^{-5} Pa\ s} = 2.61\ x\ 10^5$$

Laminar Predictions

Predictions for the thicknesses for laminar flow were obtained using the Blasius solution. For the Boundary layer thickness,

$$\delta = \frac{5.0x}{\sqrt{Re_x}} = \frac{5.0(0.165m)}{\sqrt{2.61\ x\ 10^5}} = 1.614 \text{ mm.}$$

The displacement thickness is calculated,

$$\delta^* = \frac{1.721x}{\sqrt{Re_x}} = \frac{1.721(0.165m)}{\sqrt{2.61\ x\ 10^5}} = 0.556 \text{ mm.}$$

Finally, the momentum thickness is calculated

$$\Theta = \frac{0.664x}{\sqrt{Re_x}} = \frac{0.664(0.165m)}{\sqrt{2.61\ x\ 10^5}} = 0.214 \text{ mm.}$$

Turbulent Predictions

The thicknesses for turbulent flow are calculated from empirically derived formulas that also depend on the Reynolds number. The displacement thickness is calculated using the following correlation,

$$\delta = \frac{0.0463x}{Re_x^{1/5}} = \frac{0.370(0.165m)}{(2.61\ x\ 10^5)^{1/5}} = 5.038\ mm.$$

The correlation for the displacement thickness is used,

$$\delta^* = \frac{0.0463x}{Re_x^{1/5}} = \frac{0.0462(0.165m)}{(2.61\ x\ 10^5)^{1/5}} = 0.630\ mm.$$

Finally, the following correlation is used to determine the momentum thickness,

$$\Theta = \frac{0.0360x}{Re_x^{1/5}} = \frac{0.0360(0.165m)}{(2.61\ x\ 10^5)^{1/5}} = 0.490\ mm.$$

ENGINEERING DATA SHEET	Sheet No.	D – 1/2	Date: 03/02/26
Attachment D: Sample Calculation for Drag Force and Drag Force Coefficient	Prepared	GC	03/05/26
	Checked	GL	03/05/26
	Approved	LMS	03/05/26

Drag Force Calculation

The drag force acting on the flat plate was calculated using the integral relation for a boundary layer over a flat plate:

$$F_D = \rho\, b\, U^2\, \Theta$$

where:

- ρ= fluid density
- b= plate width (perpendicular to flow direction)
- U= free-stream velocity
- Θ= momentum thickness

This relation follows directly from the momentum integral equation and represents the loss of momentum flux in the boundary layer due to viscous shear at the wall.

Experimental Values

From measured laboratory data:

- Free-stream velocity:
$$U = 23.556 \text{ m/s}$$
- Air density (calculated from ambient conditions):
$$\rho = 1.202 \text{ kg/m}^3$$
- Plate width:
$$b = 0.30 \text{ m}$$
- Momentum thickness (from trapezoidal integration):
$$\Theta = 0.167 \text{ mm} = 1.67 \times 10^{-4} \text{ m}$$

Step 1: Square the Free-Stream Velocity

$$U^2 = (23.556)^2 = 554.87 \text{ m}^2/\text{s}^2$$

Step 2: Substitute into Drag Equation

$$F_D = (1.202)(0.30)(554.87)(1.67 \times 10^{-4}) = 0.0334\ N$$

Final Drag Force

$$\boxed{F_D = 3.34 \times 10^{-2}\ \text{N}}$$

Drag Coefficient Calculation

The dimensionless drag coefficient was calculated using:

$$C_D = \frac{2\Theta}{L}$$

where $L$is the streamwise distance from the leading edge to the measurement location.

Using:

$$L = 0.165\ \text{m}$$
$$C_D = \frac{2(1.67 \times 10^{-4})}{0.165}$$
$$C_D = 0.00202$$

Final Drag Coefficient

$$\boxed{C_D = 0.00202}$$

ENGINEERING DATA SHEET	Sheet No.	E – 1/3	Date: 03/02/26
Attachment E: Quantitative Comparison of Experimental Results to Laminar and Turbulent Predictions	Prepared	GC	03/05/26
	Checked	GL	03/05/26
	Approved	LMS	03/05/26

To determine whether the measured boundary layer was laminar, turbulent, or transitional, the experimentally determined boundary layer thickness and drag coefficient were compared to theoretical laminar and turbulent flat-plate predictions.

1. Reynolds Number at Measurement Location

The Reynolds number based on distance from the leading edge is:

$$Re_x = \frac{Ux}{\nu}$$

Using:

- $U = 23.556$ m/s
- $x = 0.165$ m
- $\nu = 1.46 \times 10^{-5}$ m^2/s

$$Re_x = \frac{(23.556)(0.165)}{1.79 \times 10^{-5}}$$

$$Re_x = 2.61 \times 10^5$$

Since transition for a smooth flat plate typically occurs near $Re_x \approx 5 \times 10^5$, the measured Reynolds number is below the critical transition value, indicating that the boundary layer should be laminar at this location.

2. Comparison of Boundary Layer Thickness

Experimental:

$$\delta_{exp} = 1.645 \text{ mm}$$

Laminar Prediction (Blasius solution):

$$\frac{\delta}{x} = \frac{5.0}{\sqrt{Re_x}}$$

$$\delta_{lam} = \frac{5.0(0.165)}{\sqrt{2.61 \times 10^5}}$$

$$\delta_{lam} = 1.614 \text{ mm}$$

Turbulent Prediction:

$$\frac{\delta}{x} = 0.37 Re_x^{-1/5}$$

$$\delta_{turb} = 0.37(0.165)(2.66 \times 10^5)^{-1/5}$$

$$\delta_{turb} = 5.038 \text{ mm}$$

Percent Difference

Laminar vs Experimental:

$$\% \, error = \frac{1.645 - 1.614}{1.614} \times 100$$
$$= 1.9\%$$

Turbulent vs Experimental:

$$\% \, error = \frac{5.038 - 1.645}{5.038} \times 100$$
$$= 67.3\%$$

The experimental boundary layer thickness agrees within 2% of the laminar prediction, but differs by about 67% from the turbulent prediction.

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3. Comparison of Drag Coefficient
Experimental:

$$C_D = 0.00202$$

Laminar Prediction:

$$C_D = \frac{1.33}{\sqrt{Re_L}}$$
$$C_{D,lam} = \frac{1.33}{\sqrt{2.66 \times 10^5}}$$
$$C_{D,lam} = 0.00258$$

Percent difference:

$$\frac{0.00258 - 0.00202}{0.00258} \times 100 = 21.7\%$$

Turbulent Prediction:

$$C_D = \frac{0.074}{Re_L^{1/5}}$$
$$C_{D,turb} = 0.00606$$

This differs from the experimental value by approximately:

$$67\%$$